

Department of Computer Science
University of Pretoria

Programming Languages
COS 333

Practical 6: Python

13 May, 2026

1 Objectives

This practical aims to achieve the following general learning objective:

- To gain and consolidate some experience writing programs in an imperative scripting language, namely Python;
- To consolidate a variety of basic concepts related to imperative programming languages and scripting languages, as presented in the prescribed textbook for this course.

2 Plagiarism Policy

Plagiarism is a serious form of academic misconduct. It involves both appropriating someone else's work and passing it off as one's own work afterwards. Thus, you commit plagiarism when you present someone else's written or creative work (words, images, ideas, opinions, discoveries, artwork, music, recordings, computer-generated work, etc.) as your own. Note that using material produced in whole or part by an AI-based tool (such as ChatGPT) also constitutes plagiarism. Only hand in your own original work. Indicate precisely and accurately when you have used information provided by someone else. Referencing must be done in accordance with a recognised system. Indicate whether you have downloaded information from the Internet. For more details, visit the library's website: <http://www.library.up.ac.za/plagiarism/>.

3 Submission Instructions

Upload your practical-related source code files to the appropriate assignment upload slot on the ClickUP course page. For your Python submission you must implement and submit a single file named `u99999999.py`, where `99999999` is your student number. Multiple uploads are allowed, but only the last one will be marked. The submission deadline is **Friday, 22 May 2026, at 12:00** (note that marking will take place in the weeks of 18 May 2026 and 25 May 2026, so you are welcome to have the practical marked before the deadline).

4 Background Information

For this practical, you will be writing programs in Python 3.14. You will have to investigate this language in terms of its support for different programming language concepts. To do this, you will have to write a short program to demonstrate how the language handles the concept under consideration. This program will not be long, but will demonstrate the concepts adequately, and should allow you to understand the language's support (or lack of support) for the features in question.

Your Python 3.14 program should run using its command line interpreter interface. Be sure to check the version of the interpreter you are using. This is important because in some cases later versions of language specifications are not backward compatible with earlier versions. Manuals (which include tutorials) Python [1] are available online.

5 Practical Tasks

For this practical, you will have to write a program in Python 3.14 to perform a processing task on the contents of a text file. Your script must execute using the command line interpreter for Python. Information on how to use the command line interpreter is available via man pages under Linux, and in online resources.

The program should receive two user-provided command line parameters. Command line parameters are included after the command to execute the Python interpreter, and are separated by spaces. For more information on command line parameters, please refer to the documentation for Python, or other online resources.

The first command line parameter is the name of an ASCII input file. The second command line parameter is a word (that is, a string of characters with no spaces). The programs must first read the specified input file. The file consists of a series of student numbers and names, separated by a comma and a space. The file has the following general format:

```
2255559, John Peter Stevens
2212191, Patricia Mary Daniels
1218999, Jack Timm
1224312, Joe Black
```

Note that names may consist of any number of first and middle names, but only one last name (each name is separated by any number of spaces). A sample input file, named `input.txt`, is provided on the course ClickUP page (note that your program must work with a similarly formatted input file with an arbitrary number of student details, and that an input file may be empty).

Your programs must then search the input file and locate all students with a last name that is alphabetically after the word provided as the second command line parameter. If no such students exist, the program should print out “None found”. If any number of students with last names alphabetically after the provided word is found, only the student numbers of all these students should be printed in numeric order on separate lines. The provided word and all last names should be treated as caseless (that is, they are treated as if all letters are either all lowercase or all uppercase). There are, of course, several ways in which this operation can be performed, and you are free to use whichever approach you prefer. One possible approach to extracting the student number and last name on each line is to use a regular expression.

Given the above input file, assume that the second parameter is `sabre`. There are two students with two last names alphabetically after `sabre`, `John Peter Stevens` and `Jack Timm`. The program will then print the student numbers in the order `1218999` and `2255559`, because the former is the student number of `Jack Timm` and the latter is the student number of `John Peter Stevens`. Note that the order in which the names appear in the file is irrelevant. The only thing deciding order is the numeric relationships between student numbers.

Note that you may use any features that are part of the basic Python 3.14 language installation itself. **You may not use any third party libraries or extensions.**

6 Marking

The implementation will count a total of 5 marks. Submit the Python implementation to the appropriate assignment upload slot. Do not upload any additional files other than your source code. Both the implementation and the correct execution of the program will be taken into account. Your program code will be assessed during the practical sessions in the weeks of 18 May 2026 and 25 May 2026. Note that there is only one practical session in the week of 25 May 2026 (on the Monday).

References

- [1] Python Software Foundation. Python 3.14.5 documentation, 2026. Access at <https://docs.python.org/3/>.